TD2 report : Economic networks

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1. Exercise 1 : Demand curve and network effects (see notebook section Exercise 1 and Figure 1)

Initially, the reservation price is defined as $r(x) = 1 - x$ (where x is the location of the customer, please refer to the demonstration in the previous report). However, this does not account for social influence effects. To incorporate these, we introduce a social effect function $f(z) = z$, which depends on the number of adopters z. Assuming that the proportion of adopters is accurate, we set $x = z$. Consequently, the complete consumption value becomes :

$$r(x)f(x) = x(1 - x) \tag{1}$$

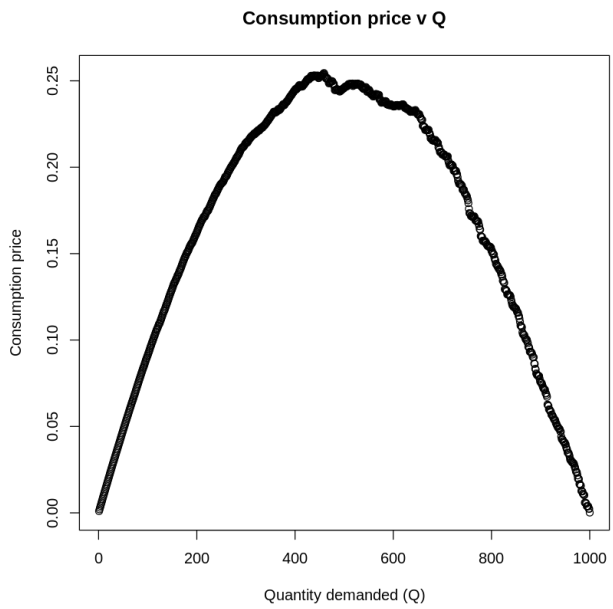

Figure 1: Consumption price v Quantity demanded (Q).

We observe a typical graph in the shape of an inverted bell curve.

2. Exercise 2 : Diffusion of innovation on social networks (see notebook section Exercise 2)

In this assignment, we will estimate the effect of social networks on individual adoption decisions of an online messenger. Therefore, we study two databases :

- `messenger_adoption.csv` : a self-reported demographic information, including gender (variable `sex` = 1 if male, 0 if female), `age` (in years), and the `time` elapsed between the public launch of the messenger and the adoption time, measured in days. The study continued for 300 days.

- `users_socnet.csv`: An edgelist corresponding to the social network of users, which presumably remained the same throughout the study.

2.1. Non-parametric model using Kaplan Meier estimator (see notebook section Exercise 2 Question 1)

The Kaplan-Meier estimator is a non-parametric method used to estimate the survival function from a dataset. For practical reasons, instead of plotting the Kaplan-Meier curve directly, we will focus on $ECDF = 1 - \text{survival function}$, which is strictly equivalent but more convenient for this task. Specifically, we will use two methods to estimate it for each subset of the studied population : by calculating the Empirical Cumulative Distribution Function $ECDF$ and by using the `survival` package in `R` to plot $1 - \text{survival function}$.

2.1.1. Kaplan Meier estimator on the entire population

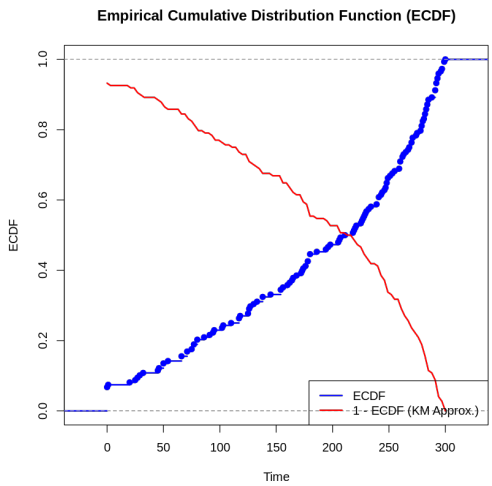

Figure 2: $ECDF = 1 - \text{survival function}$.

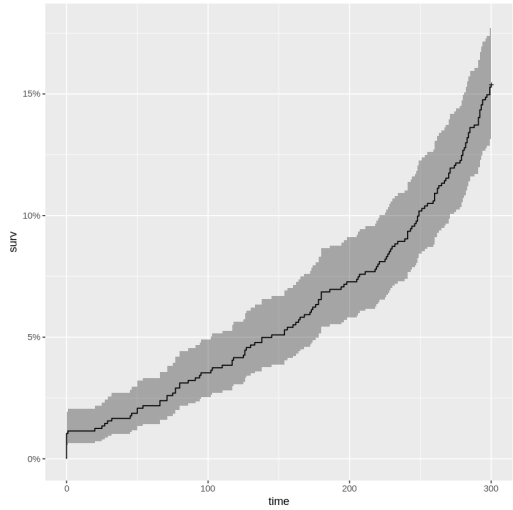

Figure 3: $ECDF = 1 - \text{survival function}$ computed with `survival` package.

We obtain the same figures with both methods. The function increases, indicating decreasing amount of survivors. In other words, more and more people are adopting the messenger regularly over time, with the same rate of registration.

2.1.2. Kaplan Meier estimator by gender

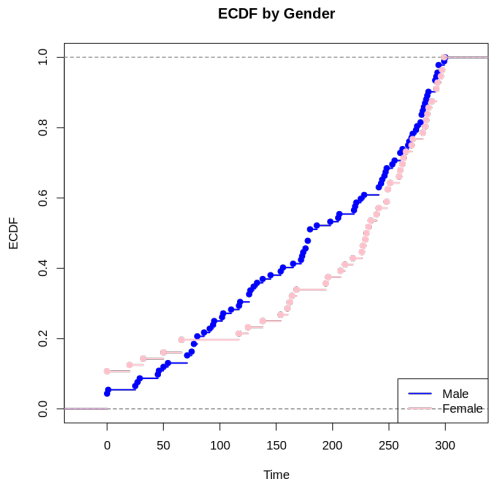

Figure 4: $ECDF = 1-$survival function.

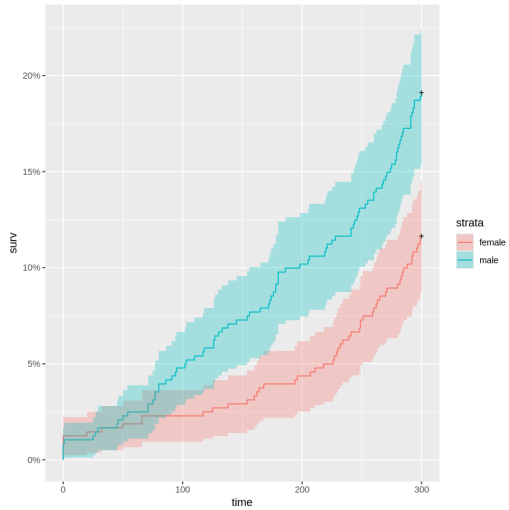

Figure 5: $ECDF = 1-$survival function computed with `survival` package.

We do not obtain identical results from both methods. This difference may be attributed to the presence of censored data, as we only know whether individuals have adopted the messenger or not, without further information regarding the reasons behind their decision. The `survival` package in `R` accounts for potential biases, such as the possibility that some individuals provided false information.

We will keep Figure 5. Early adopters are represented across both genders. Upon examining the function, we observe that the survival probability for men is lower than that for women towards the end of the observation period, indicating that men tend to adopt the messenger at a faster rate and more frequently than women. This suggests that gender influences the timing of adoption.

2.1.3. Kaplan Meier estimator by age

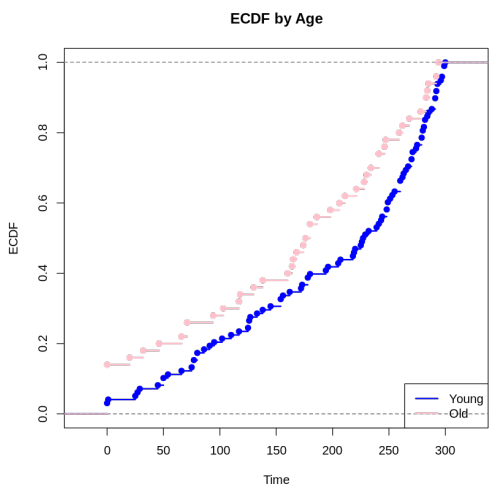

Figure 6: $ECDF = 1-$survival function.

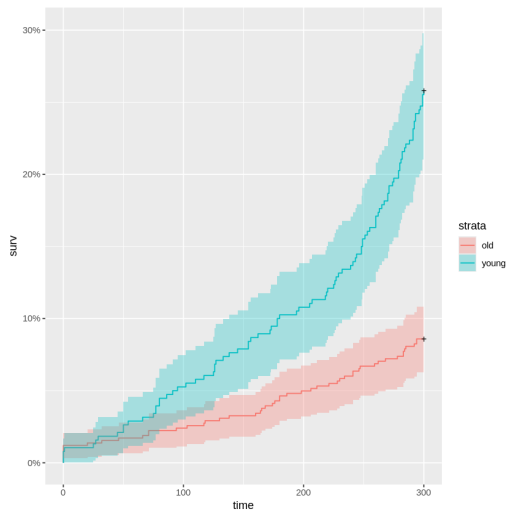

Figure 7: $ECDF = 1-$survival function computed with `survival` package.

Once again, we obtain two different figures likely for the same reasons and we will keep Figure 7. Younger users tend to adopt the messenger faster than older users, as evidenced by the steeper $ECDF$

curve for the younger group. This is expected, as younger users generally adopt new technologies at a faster rate.

2.2. Cox model of proportional hazards with time-independent covariates (see notebook section Exercise 2 Question 2)

In fact, the Kaplan-Meier estimator is a descriptive model and does not provide any explanatory power. To begin explaining the observed phenomena, we will use the Cox proportional hazards model with time-independent features.

Table 1: Cox Proportional Hazards Model

	Dependent variable:
	Cox prop. hazards Model 1
Sex	0.789***
	(0.172)
Age	−0.007
	(0.070)
Age^2	−0.001
	(0.001)
Degree Centrality	0.008**
	(0.004)
Observations	962
R^2	0.119
Max. Possible R^2	0.876
Wald Test	113.680*** (df = 4)
LR Test	121.933*** (df = 4)
Score (Logrank) Test	146.102*** (df = 4)
Note:	*p<0.1; **p<0.05; ***p<0.01

The Cox Proportional Hazard Model analysis revealed two statistically significant results with low p-values :

- Gender was found to influence the hazard rate of adoption, with males showing a higher hazard rate ($HR = 0.789$), indicating that men adopt the messenger more quickly than women.
- Degree centrality, another variable included in the model, does not seem to have any significant effect ($HR = 0.008$).

Age also exhibited a negative effect on the hazard rate ($HR = -0.007$), suggesting that older users are slower to adopt the messenger. However, this effect was only marginally significant. To investigate potential non-linear effects of age, a quadratic term (Age^2) ($HR = -0.001$) was added to the model. The results showed that the quadratic term was not significant, indicating no strong evidence of a non-linear relationship between age and adoption behavior. (Age^2) also enables us to study the very young and the very old ones.

2.3. Cox model of proportional hazards with time-independent covariates and social influence on the adoption decision (see notebook section Exercise 2 Question 3)

The total number of friends an individual has is less relevant in this context. Instead, it is more informative to consider the number of friends already using the messenger at the time of registration. Therefore, we rerun the Cox Proportional Hazard Model, incorporating the number of friends using the messenger at the time of adoption, and present the results.

Table 2: Cox Proportional Hazards Models with Social Influence

	Dependent variable:	
	time	
	Model 1	Model 2
	(1)	(2)
Sex	0.789***	0.628***
	(0.172)	(0.173)
Age	−0.007	−0.388***
	(0.070)	(0.052)
Age^2	−0.001	0.003***
	(0.001)	(0.001)
Degree Centrality	0.008**	
	(0.004)	
Num Adopting Neighbors		−0.156***
		(0.016)
Observations	962	962
R^2	0.119	0.217
Max. Possible R^2	0.876	0.876
Wald Test (df = 4)	113.680***	262.400***
LR Test (df = 4)	121.933***	235.907***
Score (Logrank) Test (df = 4)	146.102***	339.391***
Note:	*p<0.1; **p<0.05; ***p<0.01	

All results are statistically significant with low p-values. The previous observation regarding gender remains consistent. Age continues to have a negative effect on the hazard rate ($HR = -0.388$), indicating that older users adopt more slowly, and this result is now more significant. The quadratic term for age (Age^2) remains insignificant, suggesting no strong non-linear relationship between age and adoption.

Regarding social influence, the number of adopting neighbors has a significant negative effect on survival time ($HR = -0.156$). This is a counterintuitive result, as it suggests a phenomenon of 'social saturation': the higher the number of adopting neighbors, the lower the instantaneous probability of adoption. This may indicate that the remaining adopters are more resistant to adoption or that a selection effect is at play, where those most likely to adopt have already done so earlier in the process.

This new model only includes statistically significant results. Additionally, the R^2 value increases, which suggests that this model is more consistent and explanatory compared to the previous one.

2.4. Conclusion

On the one hand we have **demographic factors** : males are more likely to adopt the service earlier than females. Age negatively affects adoption, with older users adopting more slowly. On the other hand we have **social influence** : peer effects, measured by the number of adopting neighbors, exhibit a counterintuitive pattern. While an increase in adopting neighbors initially accelerates adoption timing, a phenomenon of 'social saturation' may occur, where the probability of adoption decreases among the remaining users, who are either more resistant or part of a selection effect.

3. Appendix (see notebook section Exercise 2 Question 1)

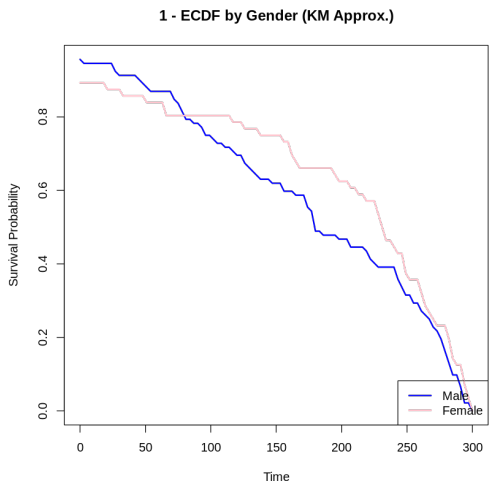

Figure 8: Survival probability by gender (KM approx).

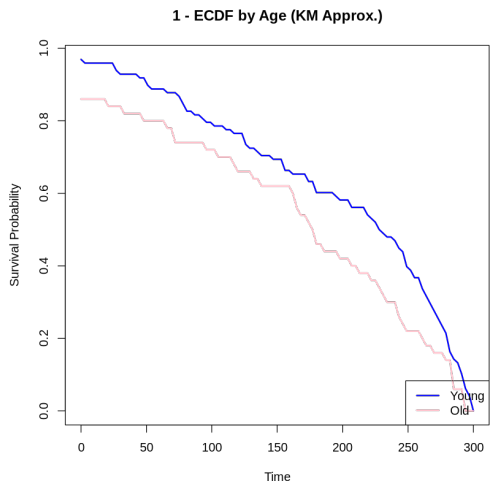

Figure 9: Survival probability by age (KM approx).